

Cover Letter

Dear Editor,

I would like to submit a manuscript titled “PACE: Probabilistic Adaptive Contention Control for Non-Primary Channel Access in IEEE 802.11bn” by Taewon Song and Yeunwoong Kyung for consideration in the IEEE Transactions on Wireless Communications.

In this manuscript, we propose PACE, a distributed probabilistic contention control scheme for non-primary channel access (NPCA) that lets stations track the time-varying fair-share transmission probability of a depleting transmission window without central coordination. The core problem is that the IEEE 802.11bn draft has stations reuse binary exponential backoff on the non-primary channel, yet that rule was designed for open-ended contention: within a bounded window, the doubled backoff frequently exceeds the time remaining, so stations fall silent while the channel idles. PACE addresses this by combining a deadline-scaled initialization, a multiplicative feedback rule in which stations copy the successful sender’s probability on solo receptions, and a self-exclusion rule that silences stations whose frame no longer fits the remaining window. A finite-horizon stochastic approximation analysis of the feedback rule yields an inverse-square-root scaling law for the adaptation step in the window length, and the resulting window-scaled variant, PACE-dynamic, derives every design parameter of the scheme from the one quantity the NPCA procedure already signals. Under IEEE 802.11 standard timing on a channel shared with native stations, PACE keeps the total useful airtime within 3% of a fair-share reference under RTS/CTS-protected access, exceeds the standard-compliant NPCA baseline by up to 61% there and by up to 4.1 times under basic access, and holds the visiting stations’ airtime at or below their population-proportional share, with PACE-dynamic preserving this proportionality across visit durations where a fixed coefficient drifts past it. An ablation study confirms that the constituent mechanisms are necessary for these gains.

We believe this work fits the scope of the IEEE Transactions on Wireless Communications in the design and analysis of wireless access protocols: the analytical contribution characterizes distributed contention control as a finite-horizon stochastic approximation over the bounded NPCA window, and the protocol contribution is standard-compatible channel access for next-generation Wi-Fi in dense overlapping-BSS deployments.

Both authors have reviewed and approved the manuscript. This manuscript has not been published in any journal and is not under consideration for publication elsewhere.

Sincerely yours,

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